

Cyber Security Architecture - My Family Setup 27/Sep/2026

HLD

Self-Manage and monitor cyber security solutions as a hobby for my family devices and network. Everything here is working in production and rarely is there an issue, except whitelists from time to time. Great experience and performance with no major complaints.

Account on social media

<https://www.linkedin.com/in/nir-roitman>

<https://github.com/dimaoudegrim>

Computers

For the family

- Checkpoint harmony endpoint.
 - URLF.
 - Anti bot.
 - Anti ransomware.
 - Behavioral.
 - Forensic.
 - Firewall.
 - Content disarm and reconstruct CDR.
 - Sandbox
 - Anti phishing.
 - DNS inspection
 - Anti malware
- 0patch for virtual patching – patching before vendors update release.
- Manageengine patch management on auto update.
- Opswat patch management on auto update (updating apps that manageengine miss).
- Zoho assist – for me to remote control.
- Browsers:

- All Browser has managed DNS firewall over https with separate profiles per person. For lowering risk and blocking ads.
- Managed allow list and sources of addons for all browsers.
- After exit, all cookies, history, custom site configuration and more are erased.
- Wazuh Agent.
 - Virustotal check for selected directories based on file hash.
 - File integrity monitoring.
 - Send Windows logs and more to siem
- Roboform password manager (will be replaced in the future)
- Bitdefender browser addon for anti-phishing.
- Ublock origin for ad blocking.
- About 168 custom registry settings for cis benchmark.
- Additional About 176 custom registry tweaks for privacy, performance, removing AI, removing ads, removing telemetry and more.
- Custom scripts to remove about 30 selected bloatwares from windows (Other were removed manually).
- Microsoft defender periodic scanning (No real time).
- Commercial vpn for selected members.

Additional For me:

- Full disk encryption.
- Block Unknown external keyboards.
- Fingerprint reader.
- Yubikey as hardware key and passkeys.
- Keepassxc.
- Customized browsers.
 - Firefox with about 155 custom settings for security, privacy and performance.
 - DNS firewall DNS over HTTPS with separated profile.

Phones

For all the family

- Up to 93.33 percent of cis benchmark for android.
- Checkpoint harmony mobile MTD.
 - URLF.
 - Anti malware
 - sandbox
- DNS Firewall.
- VPN auto-connect on untrusted network.
- Truecaller premium for family.

Additional For me:

- Secure folder for High value applications.
- MDM to separate work profile for less Important applications.
- Twingate – for connecting to Local AI.

Network

For the family

- Sophos Firewalls.
 - URLF.
 - IPS.
 - 2 Anti malware engines.
 - Sandbox.
 - Sophos active threat protection.
 - Custom IOC Feeds with more than 200,000 objects.
 - DPI and also drop weak ciphers, old ssl, compressed ssl and other certificates.
 - Layer 2 total client Isolation (On access point wifi access)

- Layer 3 block most lan to lan (on parents firewalls, not mine because of splunk lab).
- Block wan to lan.
- Block outbound risky ports.
- Connected to DNS Firewall via DNS server on AWS.
- Geo blocking also to outbound.
- MFA for login from LAN.
- DNS Firewall – NextDNS.
 - Threat intelligence blocking feeds.
 - AI-Driven Threat Detection.
 - Google Safe Browsing.
 - Cryptojacking Protection.
 - IDN Homograph Attacks Protection.
 - Typosquatting Protection.
 - Domain Generation Algorithms (DGAs) Protection.
 - Block Newly Registered Domains (NRDs).
 - Block Dynamic DNS Hostnames.
 - Block Parked Domains.
 - Block many risky Top-Level Domains (TLDs)
 - Ad blocking
 - Tracking protection.
 - Block free hosting domains
 - Block Data Drop Services
 - Block Residential Hosting
 - Block Decentralized Web Gateways
 - block Child Sexual Abuse Material (CSAM)
 - Support dnssec.
 - Used with dns over tls or https.
- DNS server on AWS – Adguard home – connected to dns firewall.

- Threat intelligence feeds.
- Ad blocking.
- Tracking protection.

Additional For me:

- Checkpoint SMB Firewall.
- Twingate ZTNA.

Email

For the family

- Passkeys login.
- MFA.

Additional For me

- Yubikey login
- Advance Protection Program
- Check point email protection for 1 Mailbox on google workspace with a domain.
 - Dnssec
 - Spf
 - Mta-sts
 - Dmarc
 - Dkim
 - caa

Monitoring

- Wazuh SIEM - located on EC2 AWS.
 - More than 550 custom rules.
 - Alerting immediately to Slack on my phone.
 - All windows computers have agents on them.
 - All servers have agents on them.
 - The Firewalls gets monitored.
 - Checkpoint EDR monitored on pc and server.
 - Checkpoint harmony mobile monitoring.
 - Checkpoint Email monitoring
 - Virustotal on windows pc by agent.
 - File integrity monitoring.
 - Aws monitoring.
 - Cis benchmark checks.
 - Website monitoring for failures and ddos.
 - Additional services are monitored.
- Dark web monitoring
 - Nordvpn darkweb monitoring
 - Roboform – part of their dark web monitoring.
- Scout suite - AWS misconfiguration checks – Manual checks.
- Guardduty – For AWS.

Servers

- Wazuh agent installed.
- Check point harmony endpoint EDR.
- Custom security groups on ec2.
- UFW
- Auditd.
- Guarddduty.
- Login with certs from specific whitelisted IP's.
- No root login and additional hardening.
- Monthly backup. Additional quarterly backup to different regions.
- All disks are encrypted.
- Scout suite - AWS misconfiguration checks – Manual checks.
- Frequent apt updates.
- Ubuntu pro on latest version.
- Cloudtrail monitoring. S3 not public, can read access from single IP.

Website

Squarespace and bunny cdn

- CDN
- DNS
 - Dnssec
 - Spf
 - Mta-sts
 - Dmarc
 - Dkim
 - Caa
- Geo blocking.
- Rate limit.
- WAF.
- Anti bot.
- DDOS protection.
- Major access lists
 - Block VPN.
 - Block traffic from datacenters.
 - Block TOR
 - Block known BOTS.
 - Abuseipdb and other known feeds.
- Upload scanning against malwares and CSAM.
- Domain hold.
- Whois privacy.
- Block post requests.
- Request and response body limit.
- Real-time threat intelligence.

Backups

- Fast Large capacity Disk on keys up to 1TB each.
- Multiple Hard disks.
- Onedrive (6x1TB).
- Family Google storage with combined 5TB shared with separate allocation per person.
- Servers' snapshots on multiple separated regions.

AI

- Local AI Pipeline with private and secure connection between Desktop to remote phones and laptop.

The pipeline:

1. On a Desktop with 16 GB Vram (AMD) I have LM studio or Lmstudio bionic with Gemma 4 26b a4b qat. It act as a server with exposed API to the local network.
 2. On a raspberry pi on the local network I have a docker with open webui on it that connected to the API.
 3. For https I have on the raspberry pi a component called caddy.
 4. The open webui is set to enable web search with a toggle on the chat itself. The interface is set to temporary chats without saving chats. The model set to not remember memories or check older chats. shell usage disabled.
 5. On the raspberry pi was installed Twingate connector as ZTNA. On Twingate I set a resource with specific port and alias for the internal domain name.
 6. On Android phone I've set inside a secure folder the twingate app, so it won't interfere with the tunneling protocol I have on the standard profile (Check point harmony mobile MTD).
 7. I've installed Caddy generated ca cert on the secure folder. I've enabled the debug secret menu on Firefox on the secure folder, so I would be able to set Firefox to use third party CA certificates.
 8. Connecting to Twingate via fingerprint and from there to the https internal site I've set of open webui.
- Perplexity (For myself)
 - Gemini AI PRO family plan with 5TB shared storage included for all google services (for family)

- Lm studio Bionic - Local AI with gui and internet capabilities (shell usage disabled) with Lm link for use of main computer power by remote laptop.
- Lm studio – Local AI with GUI.

SPLUNK Home lab, not production, as a testing environment

Splunk Enterprise clusters with ES, as a testing environment for my enterprise clients.

Build on Vmware workstation pro as VM'S (16 VM'S) on two old computers that most of the time turned off for electricity saving. But can be turned remotely.

- 12 VM's in clusters mode.
 - Cluster manager.
 - Deployer
 - 3 Indexers in clusters mode and multi-site and 3 level indexes replications.
 - 3 Search head with Enterprise security v8.3 in a cluster mode and multi-site
 - License manager with monitoring console
 - Deployment server
 - SC4S
 - Heavy forwarder
 - Linux VM for sending logs by UF, and also syslog by Rsyslog to SC4S.
 - Windows VM for sending logs by UF, and also Syslog to SC4S by solarwinds event forwarder -> Kiwi Syslog server -> SC4S -> Splunk (Mimic air-gapped environment).
- Custom architecture for the implementation.
 - Using Raspberry pi on the same network for using wake on lan to turn on the computers (hosts).
 - Each VM gets a static IP by DHCP from check point firewall.
 - Using Twingate ztna connector on one of the VM for login to rest of the VM.

- Custom DNS records on the DNS firewall for using local hostnames for the VM's.
- Self-signed tls certificates.
- Each Linux VM can't use ROOT login, and login done by CERT key.
- The Checkpoint firewall has a always turned on switch button, in case of power drop.
- Additional servers for the splunk cluster:
 - MISP – was needed for connecting misp to threat intelligence module.
 - Wazuh – needed for testing sending to multiple sources from appliances that can send to just one dest. while one of the sources isn't splunk SIEM.

Previously used, currently not activated

- pentester[.]com (temporary canceled because of duplication and issue with data broker removals that not fully completed because of lack of supporting local laws).
 - Dark web monitoring for breached credentials.
 - data broker removal
 - osint intel.
- Yara detection and auto remove IOC on pc's (Issue to maintain remotely).

IT OTHER – no security related

For lowering risks from piracy and freeware/greyware and improve user experience.

- Microsoft office 356 family plan.
- Spotify
- Netflix/Disney plus
- Kartina.tv
- Youtube premium family plan (no ads).
- Tidal
- Adobe creative cloud
- Other entertainment apps.